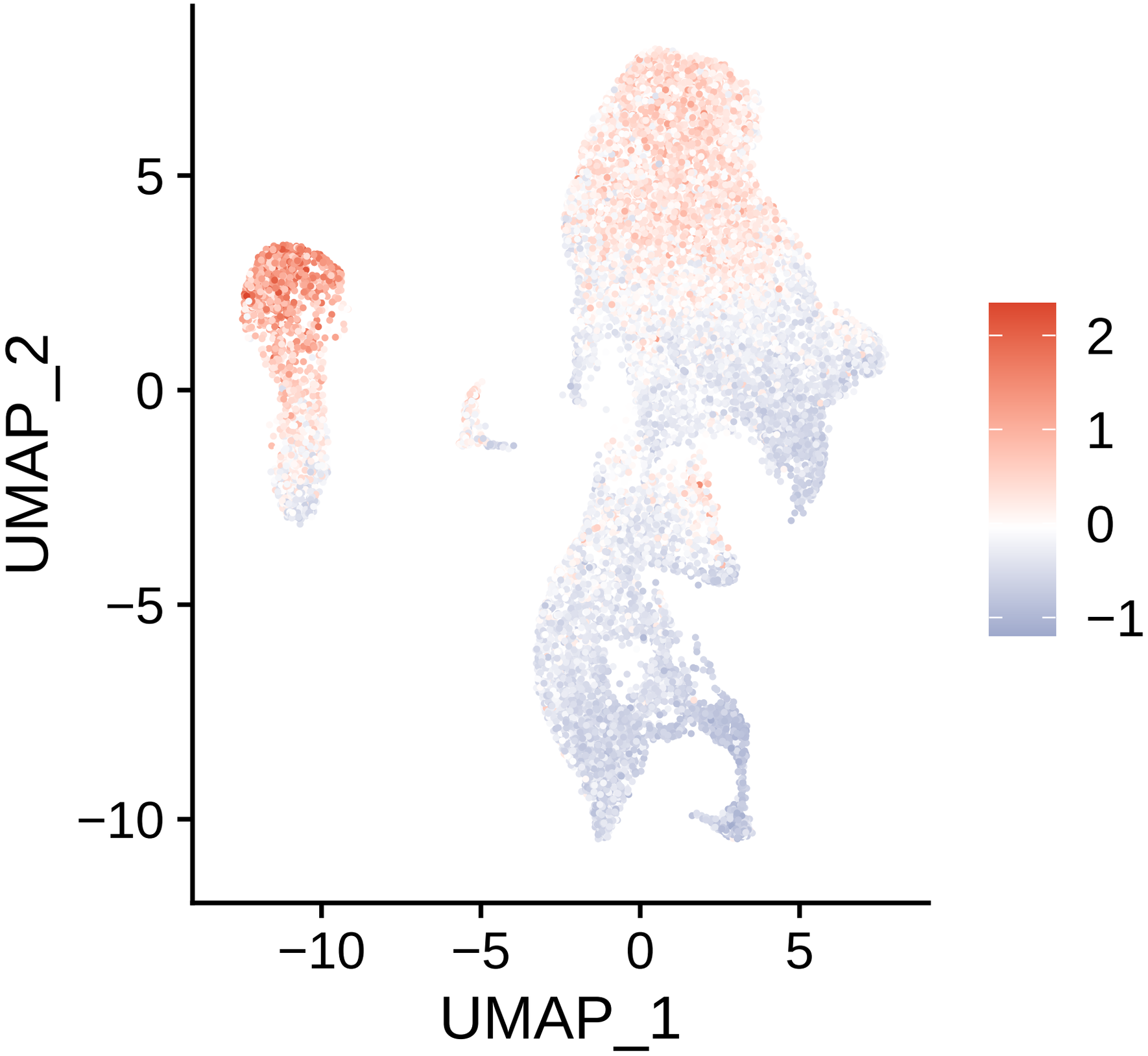LSC104_Ng2017_DOWN